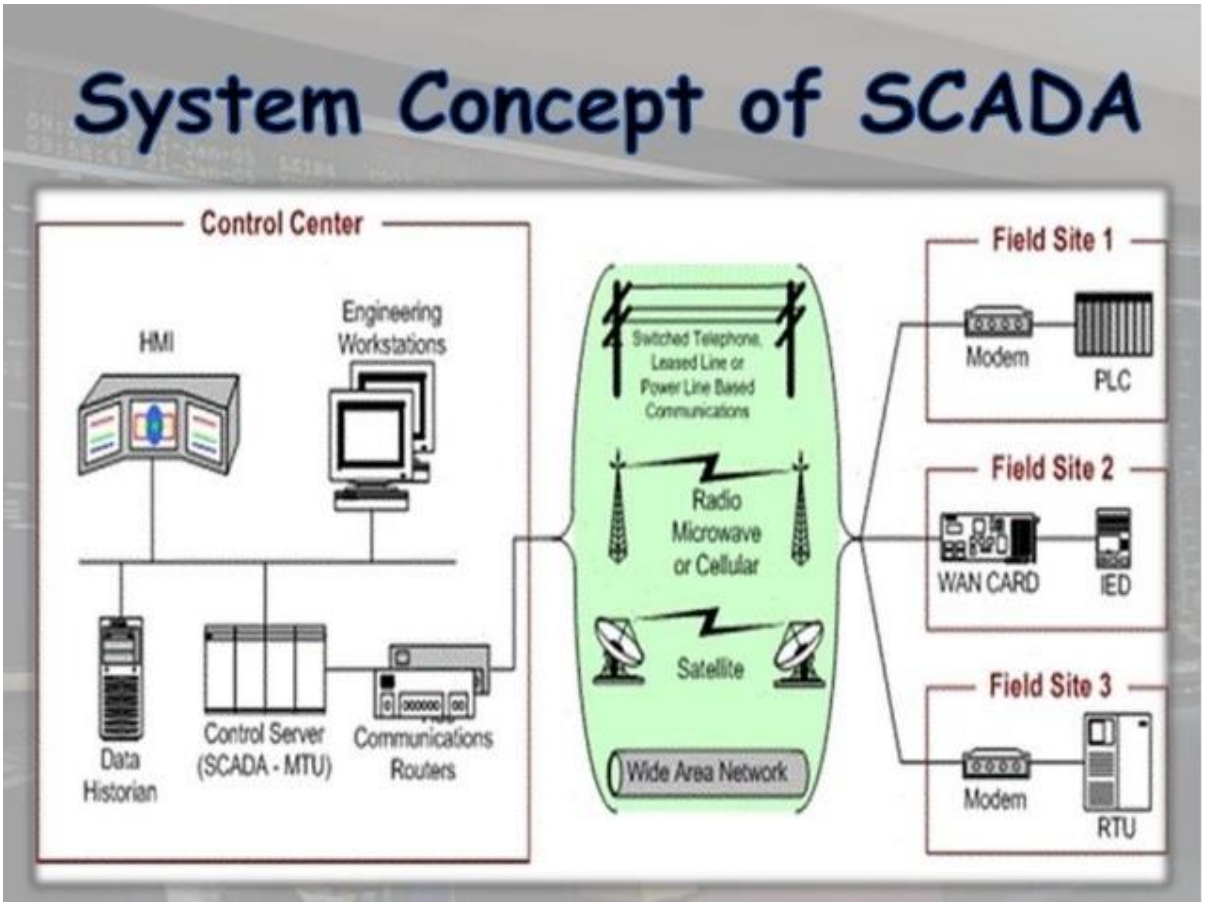


The above figure describes a general architecture and implementation concept of SCADA system, the field sites represents remote sites, in our case it represents Compact grid Substation, and the Control Center represents Tenaga's Command and Monitoring Station where MTU is the receiving end of RTUs and all other Field devices.

Keeping above in view, we have following queries regarding development, implementation and integration of RTU.

RTU:

1. What are the total number of I/O required, including Digital + Analog (in case any change is required in current model).(Already provided)
2. The current SLD / Circuit diagram for RTU installation is required. (Already provided)
3. Any change required in Power supply voltage, total number of I/O & parameters etc.
4. Protocol level parameters, Data rate, Connection medium like WIFI, Cellular, Ethernet etc.
5. Any type of local automation / configuration is required.
6. Daisy chain capability (if required).
7. Parameter List, Program Flow.

MTU / Control Center: (Tenaga Software)

1. Complete information about MTU integration, HMI / VMS Software (Existing Architecture).
2. VMS Database management, how the monitoring is done, example screens etc.
3. How the data collected is utilized and integrated with other Monitoring & control platforms (if cross platform integration is implemented).
4. How the data is being collected, its packet information, its transport/physical layer details.
5. How the collected data is processed and utilized, like report generation, Alarms indication etc.
6. A copy of VMS software if possible, or a video / Training demo.
7. How Automation is implemented, and on what parameters the automation is implemented.
8. What type of Database is implemented like SQL, NoSQL etc.
9. Access control schemes and levels like (Super Admin, Admin, User etc.)